

SNx4HC20 双路 4 输入与非门

1 特性

- 缓冲输入
- 宽工作电压范围：2V 至 6V
- 宽工作温度范围：
 - 40°C 至 +85°C
- 支持多达 10 个 LSTTL 负载的扇出
- 与 LSTTL 逻辑 IC 相比，可显著降低功耗

2 应用

- 警报/篡改检测电路
- S-R 锁存器

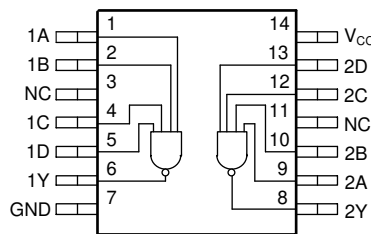
3 说明

该器件包含两个独立的 4 输入与非门。每个逻辑门以正逻辑执行布尔函数 $Y = A \bullet B \bullet C \bullet D$ 。

器件信息⁽¹⁾

器件型号	封装	封装尺寸 (标称值)
SN74HC20D	SOIC (14)	8.70mm × 3.90mm
SN74HC20N	PDIP (14)	19.30mm × 6.40mm
SN74HC20NS	SO (14)	10.20mm × 5.30mm
SN74HC20PW	TSSOP (14)	5.00mm × 4.40mm
SN54HC20J	CDIP (14)	21.30mm × 7.60mm
SN54HC20W	CFP (14)	9.20mm × 6.29mm
SN54HC20FK	LCCC (20)	8.90mm × 8.90mm

(1) 如需了解所有可用封装，请参阅数据表末尾的可订购产品附录。



功能引脚分配



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4 Revision History

注：以前版本的页码可能与当前版本的页码不同

Changes from Revision F (August 2003) to Revision G (April 2021)	Page
• 更新了整个文档中的表格、图和交叉参考的编号格式.....	1
• 更新至全新的数据表标准.....	1
• $R_{\theta JA}$ increased for the D package from 86 to 133.6°C/W, increased for the PW package from 113°C/W, and decreased for the N package from 80 to 67.2°C/W.....	4

5 Pin Configuration and Functions

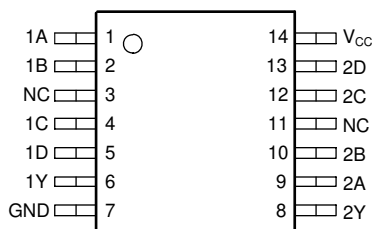


图 5-1. D, N, NS, PW, J, or W Package
14-Pin SOIC, PDIP, SO, TSSOP, CDIP, or CFP
Top View

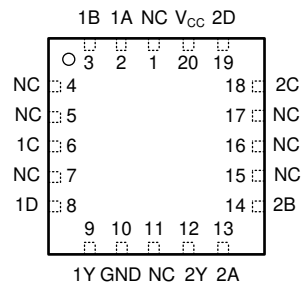


图 5-2. FK Package
20-Pin LCCC
Top View

Pin Functions

NAME	PIN		I/O	DESCRIPTION
	D, N, NS, PW, J, or W	FK		
1A	1	2	Input	Channel 1, Input A
1B	2	3	Input	Channel 1, Input B
NC	3, 11	1, 4, 5, 7, 11, 15, 16, 17	—	Not internally connected
1C	4	6	Input	Channel 1, Input C
1D	5	8	Input	Channel 1, Input D
1Y	6	9	Output	Channel 1, Output Y
GND	7	10	—	Ground
2Y	8	12	Output	Channel 2, Output Y
2A	9	13	Input	Channel 2, Input A
2B	10	14	Input	Channel 2, Input B
2C	12	18	Input	Channel 2, Input C
2D	13	19	Input	Channel 2, Input D
V _{CC}	14	20	—	Positive Supply

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

			MIN	MAX	UNIT
V _{CC}	Supply voltage		- 0.5	7	V
I _{IK}	Input clamp current ⁽²⁾	V _I < 0 or V _I > V _{CC}		±20	mA
I _{OK}	Output clamp current ⁽²⁾	V _O < 0 or V _O > V _{CC}		±20	mA
I _O	Continuous output current	V _O = 0 to V _{CC}		±25	mA
	Continuous current through V _{CC} or GND			±50	mA
T _J	Junction temperature ⁽³⁾			150	°C
T _{stg}	Storage temperature		- 65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Rating* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Condition*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

(3) Guaranteed by design.

6.2 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
V _{CC}	Supply voltage		2	5	6	V
V _{IH}	High-level input voltage	V _{CC} = 2 V	1.5			V
		V _{CC} = 4.5 V	3.15			
		V _{CC} = 6 V	4.2			
V _{IL}	Low-level input voltage	V _{CC} = 2 V			0.5	V
		V _{CC} = 4.5 V			1.35	
		V _{CC} = 6 V			1.8	
V _I	Input voltage		0		V _{CC}	V
V _O	Output voltage		0		V _{CC}	V
t _i	Input transition time	V _{CC} = 2 V			1000	ns
		V _{CC} = 4.5 V			500	
		V _{CC} = 6 V			400	
T _A	Operating free-air temperature	SN54HC00	- 55		125	°C
		SN74HC00	- 40		85	

6.3 Thermal Information

THERMAL METRIC ⁽¹⁾		SN74HC20				UNIT
		N (PDIP)	D (SOIC)	PW (TSSOP)	NS (SOP)	
		14 PINS	14 PINS	14 PINS	14 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	67.2	133.6	151.7	122.6	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	55.3	89.0	79.4	81.8	°C/W
R _{θJB}	Junction-to-board thermal resistance	46.9	89.5	94.7	83.8	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	35.3	45.5	25.2	45.4	°C/W

THERMAL METRIC ⁽¹⁾		SN74HC20				UNIT
		N (PDIP)	D (SOIC)	PW (TSSOP)	NS (SOP)	
		14 PINS	14 PINS	14 PINS	14 PINS	
Ψ_{JB}	Junction-to-board characterization parameter	46.7	89.1	94.1	83.4	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	N/A	N/A	N/A	N/A	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

6.4 Electrical Characteristics - 74

over operating free-air temperature range; typical values measured at $T_A = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER		TEST CONDITIONS		V _{CC}	Operating free-air temperature (T _A)						UNIT
					25°C			-40°C to 85°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
V _{OH}	High-level output voltage	V _I = V _{IH} or V _{IL}	I _{OH} = - 20 μA	2 V	1.9	1.998		1.9			V
				4.5 V	4.4	4.499		4.4			
				6 V	5.9	5.999		5.9			
			I _{OH} = - 4 mA	4.5 V	3.98	4.3		3.84			
				6 V	5.48	5.8		5.34			
V _{OL}	Low-level output voltage	V _I = V _{IH} or V _{IL}	I _{OL} = 20 μA	2 V		0.002	0.1			0.1	V
				4.5 V		0.001	0.1			0.1	
				6 V		0.001	0.1			0.1	
			I _{OL} = 4 mA	4.5 V		0.17	0.26			0.33	
				6 V		0.15	0.26			0.33	
I _I	Input leakage current	V _I = V _{CC} or 0		6 V		±0.1	±100			±1000	nA
I _{CC}	Supply current	V _I = V _{CC} or 0	V _I = V _{CC} or 0	6 V			2			20	μA
C _i	Input capacitance			2 V to 6 V		3	10			10	pF

6.5 Electrical Characteristics - 54

over operating free-air temperature range; typical values measured at $T_A = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER		TEST CONDITIONS		V _{CC}	Operating free-air temperature (T _A)			Operating free-air temperature (T _A)			UNIT
					25°C			– 55°C to 125°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
V _{OH}	High-level output voltage	V _I = V _{IH} or V _{IL}	I _{OH} = – 20 μA	2 V	1.9	1.998		1.9			V
				4.5 V	4.4	4.499		4.4			
				6 V	5.9	5.999		5.9			
			I _{OH} = – 4 mA	4.5 V	3.98	4.3		3.7			
			I _{OH} = – 5.2 mA	6 V	5.48	5.8		5.2			
V _{OL}	Low-level output voltage	V _I = V _{IH} or V _{IL}	I _{OL} = 20 μA	2 V		0.002	0.1			0.1	V
				4.5 V		0.001	0.1			0.1	
				6 V		0.001	0.1			0.1	
			I _{OL} = 4 mA	4.5 V		0.17	0.26			0.4	
			I _{OL} = 5.2 mA	6 V		0.15	0.26			0.4	

over operating free-air temperature range; typical values measured at $T_A = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER		TEST CONDITIONS		V _{CC}	Operating free-air temperature (T _A)			Operating free-air temperature (T _A)			UNIT
					25°C			– 55°C to 125°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
I _I	Input leakage current	V _I = V _{CC} or 0		6 V	±0.1			±1			µA
I _{CC}	Supply current	V _I = V _{CC} or 0	I _O = 0	6 V	2			40			µA
C _i	Input capacitance			2 V to 6 V	3 10			10			pF

6.6 Switching Characteristics - 74

over operating free-air temperature range (unless otherwise noted)

PARAMETER		FROM	TO	V _{CC}	Operating free-air temperature (T _A)						UNIT
					25°C			– 40°C to 85°C			
					MIN	TYP	MAX	MIN	TYP	MAX	
t _{pd}	Propagation delay	A, B, C, or D	Y	2 V	45		110	140			ns
				4.5 V	14		22	28			
				6 V	11		19	24			
t _t	Transition-time		Y	2 V	27		75	95			ns
				4.5 V	9		15	19			
				6 V	7		13	16			

6.7 Switching Characteristics - 54

over operating free-air temperature range; typical values measured at $T_A = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER		FROM	TO	TEST CONDITIONS	V _{CC}	Operating free-air temperature (T _A)			Operating free-air temperature (T _A)			UNIT
						25°C			– 55°C to 125°C			
						MIN	TYP	MAX	MIN	TYP	MAX	
t _{pd}	Propagation delay	A, B, C, or D	Y	C _L = 50 pF	2 V		45	110			140	ns
					4.5 V		9	22			28	
					6 V		8	19			24	
t _t	Transition-time		Y	C _L = 50 pF	2 V		27	75			95	ns
					4.5 V		9	15			19	
					6 V		7	13			16	

6.8 Operating Characteristics

over operating free-air temperature range; typical values measured at $T_A = 25^\circ\text{C}$ (unless otherwise noted).

PARAMETER		TEST CONDITIONS	V_{CC}	MIN	TYP	MAX	UNIT
C_{pd}	Power dissipation capacitance per gate	No load	2 V to 6 V		25		pF

6.9 Typical Characteristics

$T_A = 25^\circ\text{C}$

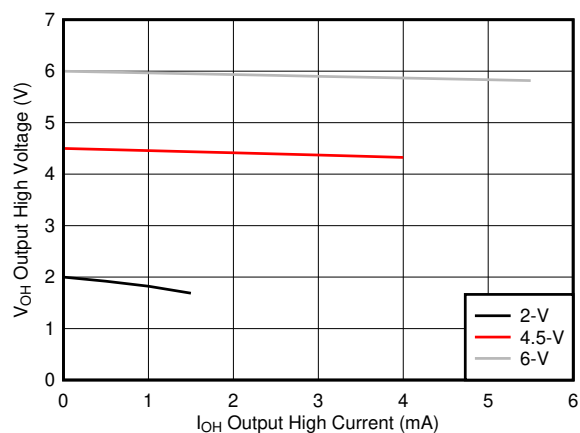


图 6-1. Typical output voltage in the high state (V_{OH})

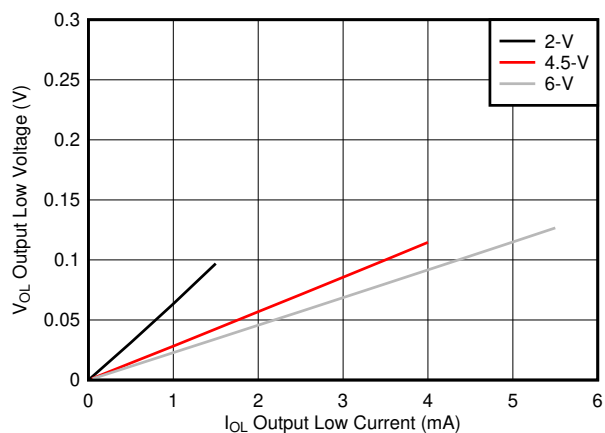
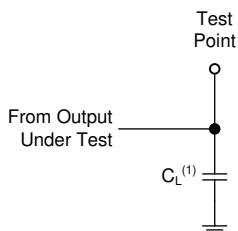


图 6-2. Typical output voltage in the low state (V_{OL})

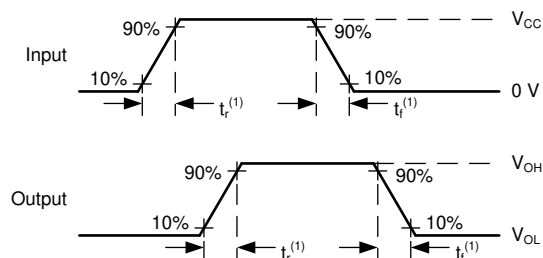
7 Parameter Measurement Information

- Phase relationships between waveforms were chosen arbitrarily. All input pulses are supplied by generators having the following characteristics: $PRR \leq 1 \text{ MHz}$, $Z_O = 50 \Omega$, $t_t < 6 \text{ ns}$.
- The outputs are measured one at a time, with one input transition per measurement.



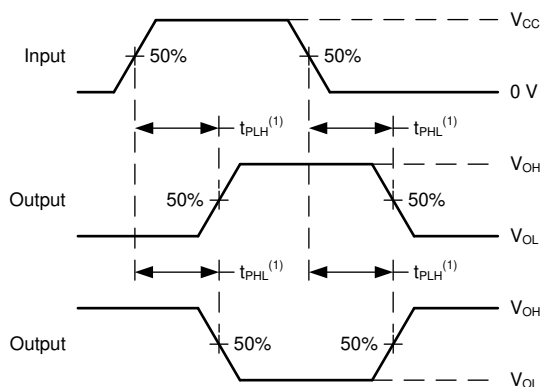
A. $C_L = 50 \text{ pF}$ and includes probe and jig capacitance.

图 7-1. Load Circuit



A. t_t is the greater of t_r and t_f .

图 7-2. Voltage Waveforms Transition Times



A. The maximum between t_{PLH} and t_{PHL} is used for t_{pd} .

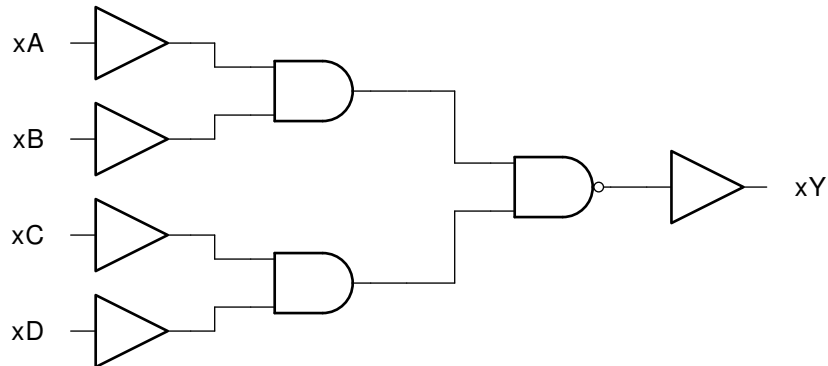
图 7-3. Voltage Waveforms Propagation Delays

8 Detailed Description

8.1 Overview

This device contains two independent 4-input NAND gates. Each gate performs the Boolean function $Y = A \cdot B \cdot C \cdot D$ in positive logic.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Balanced CMOS Push-Pull Outputs

A balanced output allows the device to sink and source similar currents. The drive capability of this device may create fast edges into light loads so routing and load conditions should be considered to prevent ringing. Additionally, the outputs of this device are capable of driving larger currents than the device can sustain without being damaged. It is important for the output power of the device to be limited to avoid damage due to over-current. The electrical and thermal limits defined in the [Absolute Maximum Ratings](#) must be followed at all times.

The SN74HC20 can drive a load with a total capacitance less than or equal to the maximum load listed in the [Switching Characteristics - 74](#) connected to a high-impedance CMOS input while still meeting all of the datasheet specifications. Larger capacitive loads can be applied, however it is not recommended to exceed the provided load value. If larger capacitive loads are required, it is recommended to add a series resistor between the output and the capacitor to limit output current to the values given in the [Absolute Maximum Ratings](#).

8.3.2 Standard CMOS Inputs

Standard CMOS inputs are high impedance and are typically modeled as a resistor from the input to ground in parallel with the input capacitance given in the [Electrical Characteristics - 74](#). The worst case resistance is calculated with the maximum input voltage, given in the [Absolute Maximum Ratings](#), and the maximum input leakage current, given in the [Electrical Characteristics - 74](#), using ohm's law ($R = V \div I$).

Signals applied to the inputs need to have fast edge rates, as defined by the input transition time in the [Recommended Operating Conditions](#) to avoid excessive current consumption and oscillations. If a slow or noisy input signal is required, a device with a Schmitt-trigger input should be used to condition the input signal prior to the standard CMOS input.

8.3.3 Clamp Diode Structure

The inputs and outputs to this device have both positive and negative clamping diodes as depicted in 图 8-1.

CAUTION

Voltages beyond the values specified in the 表 6.1 table can cause damage to the device. The recommended input and output voltage ratings may be exceeded if the input and output clamp-current ratings are observed.

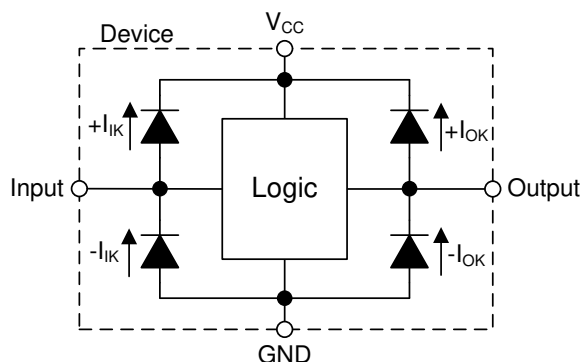


图 8-1. Electrical Placement of Clamping Diodes for Each Input and Output

8.4 Device Functional Modes

表 8-1. Function Table

INPUTS				OUTPUT
A	B	C	D	Y
H	H	H	H	L
L	X	X	X	H
X	L	X	X	H
X	X	L	X	H
X	X	X	L	H

9 Application and Implementation

备注

以下应用部分中的信息不属于 TI 器件规格的范围，TI 不担保其准确性和完整性。TI 的客户应负责确定器件是否适用于其应用。客户应验证并测试其设计，以确保系统功能。

9.1 Application Information

In this application, two 4-input NAND gates are used to create an active-low SR latch as shown in [图 9-1](#). Unused inputs can be tied together or to V_{CC} .

This device is used to drive the tamper indicator LED and provide one bit of data to the system controller. When the tamper switch outputs LOW, the output Q becomes HIGH. This output remains HIGH until the system controller addresses the event and sends a LOW signal to the R input which returns the Q output back to LOW.

9.2 Typical Application

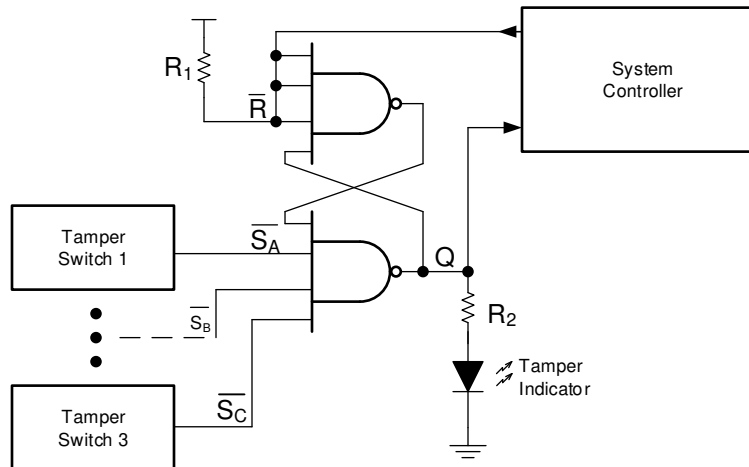


图 9-1. Typical application schematic

9.2.1 Design Requirements

9.2.1.1 Power Considerations

Ensure the desired supply voltage is within the range specified in the [Recommended Operating Conditions](#). The supply voltage sets the device's electrical characteristics as described in the [Electrical Characteristics - 74](#).

The supply must be capable of sourcing current equal to the total current to be sourced by all outputs of the SN74HC20 plus the maximum supply current, I_{CC} , listed in the [Electrical Characteristics - 74](#). The logic device can only source or sink as much current as it is provided at the supply and ground pins, respectively. Be sure not to exceed the maximum total current through GND or V_{CC} listed in the [Absolute Maximum Ratings](#).

Total power consumption can be calculated using the information provided in [CMOS Power Consumption and \$C_{pd}\$ Calculation](#).

Thermal increase can be calculated using the information provided in [Thermal Characteristics of Standard Linear and Logic \(SLL\) Packages and Devices](#).

CAUTION

The maximum junction temperature, $T_J(\text{max})$ listed in the [Absolute Maximum Ratings](#), is an *additional limitation* to prevent damage to the device. Do not violate any values listed in the [Absolute Maximum Ratings](#). These limits are provided to prevent damage to the device.

9.2.1.2 Input Considerations

Unused inputs must be terminated to either V_{CC} or ground. These can be directly terminated if the input is completely unused, or they can be connected with a pull-up or pull-down resistor if the input is to be used sometimes, but not always. A pull-up resistor is used for a default state of HIGH, and a pull-down resistor is used for a default state of LOW. The resistor size is limited by drive current of the controller, leakage current into the SN74HC20, as specified in the [Electrical Characteristics - 74](#), and the desired input transition rate. A 10-k Ω resistor value is often used due to these factors.

The SN74HC20 has standard CMOS inputs, so input signal edge rates cannot be slow. Slow input edge rates can cause oscillations and damaging shoot-through current. The recommended rates are defined in the [Recommended Operating Conditions](#).

Refer to [§ 8.3](#) for additional information regarding the inputs for this device.

9.2.1.3 Output Considerations

The positive supply voltage is used to produce the output HIGH voltage. Drawing current from the output will decrease the output voltage as specified by the V_{OH} specification in the [Electrical Characteristics - 74](#). Similarly, the ground voltage is used to produce the output LOW voltage. Sinking current into the output will increase the output voltage as specified by the V_{OL} specification in the [Electrical Characteristics - 74](#).

Unused outputs can be left floating. Do not connect outputs directly to V_{CC} or ground.

Refer to [§ 8.3](#) for additional information regarding the outputs for this device.

9.2.2 Detailed Design Procedure

1. Add a decoupling capacitor from V_{CC} to GND. The capacitor needs to be placed physically close to the device and electrically close to both the V_{CC} and GND pins. An example layout is shown in [§ 11](#).
2. Ensure the capacitive load at the output is ≤ 70 pF. This is not a hard limit, however it will ensure optimal performance. This can be accomplished by providing short, appropriately sized traces from the SN74HC20 to the receiving device.
3. Ensure the resistive load at the output is larger than $(V_{CC} / I_{O(max)}) \Omega$. This will ensure that the maximum output current from the [Absolute Maximum Ratings](#) is not violated. Most CMOS inputs have a resistive load measured in megaohms; much larger than the minimum calculated above.
4. Thermal issues are rarely a concern for logic gates, however the power consumption and thermal increase can be calculated using the steps provided in the application report, [CMOS Power Consumption and Cpd Calculation](#)

9.2.3 Application Curves

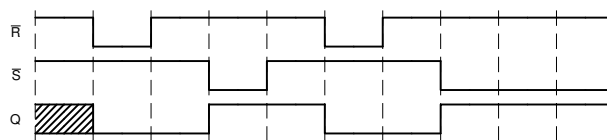


图 9-2. Typical application timing diagram

10 Power Supply Recommendations

The power supply can be any voltage between the minimum and maximum supply voltage rating located in the [§ 6.2](#). Each V_{CC} terminal should have a bypass capacitor to prevent power disturbance. A 0.1- μ F capacitor is recommended for this device. It is acceptable to parallel multiple bypass caps to reject different frequencies of noise. The 0.1- μ F and 1- μ F capacitors are commonly used in parallel. The bypass capacitor should be installed as close to the power terminal as possible for best results, as shown in [图 11-1](#).

11 Layout

11.1 Layout Guidelines

When using multiple-input and multiple-channel logic devices inputs must not ever be left floating. In many cases, functions or parts of functions of digital logic devices are unused; for example, when only two inputs of a triple-input AND gate are used. Such unused input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. All unused inputs of digital logic logic devices must be connected to a logic high or logic low voltage, as defined by the input voltage specifications, to prevent them from floating. The logic level that must be applied to any particular unused input depends on the function of the device. Generally, the inputs are tied to GND or V_{CC} , whichever makes more sense for the logic function or is more convenient.

11.2 Layout Example

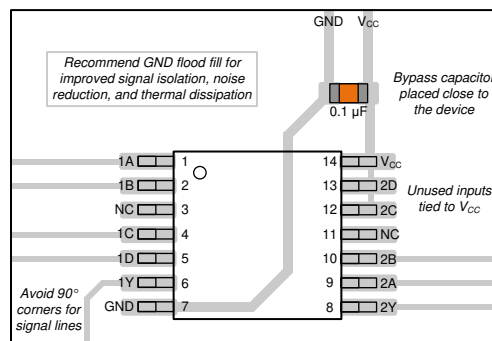


图 11-1. Example layout for the SN74HC20

12 Device and Documentation Support

12.1 Documentation Support

12.1.1 Related Documentation

For related documentation see the following:

- [HCMOS Design Considerations](#)
- [CMOS Power Consumption and CPD Calculation](#)
- [Designing with Logic](#)

12.2 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

12.3 支持资源

[TI E2E™ 支持论坛](#)是工程师的重要参考资料，可直接从专家获得快速、经过验证的解答和设计帮助。搜索现有解答或提出自己的问题可获得所需的快速设计帮助。

链接的内容由各个贡献者“按原样”提供。这些内容并不构成 TI 技术规范，并且不一定反映 TI 的观点；请参阅 TI 的《使用条款》。

12.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.

所有商标均为其各自所有者的财产。

12.5 静电放电警告



静电放电 (ESD) 会损坏这个集成电路。德州仪器 (TI) 建议通过适当的预防措施处理所有集成电路。如果不遵守正确的处理和安装程序，可能会损坏集成电路。

ESD 的损坏小至导致微小的性能降级，大至整个器件故障。精密的集成电路可能更容易受到损坏，这是因为非常细微的参数更改都可能会导致器件与其发布的规格不相符。

12.6 术语表

[TI 术语表](#) 本术语表列出并解释了术语、首字母缩略词和定义。

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
5962-8403901VCA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	5962-8403901VC A SNV54HC20J	Samples
84039012A	ACTIVE	LCCC	FK	20	55	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	84039012A SNJ54HC 20FK	Samples
8403901CA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	8403901CA SNJ54HC20J	Samples
8403901DA	ACTIVE	CFP	W	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	8403901DA SNJ54HC20W	Samples
JM38510/65003BCA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 65003BCA	Samples
M38510/65003BCA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 65003BCA	Samples
SN54HC20J	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	SN54HC20J	Samples
SN74HC20D	OBSOLETE	SOIC	D	14		TBD	Call TI	Call TI	-40 to 85	HC20	
SN74HC20DR	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU SN	Level-1-260C-UNLIM	-40 to 85	HC20	Samples
SN74HC20N	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-40 to 85	SN74HC20N	Samples
SN74HC20NSR	ACTIVE	SO	NS	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-40 to 85	HC20	Samples
SN74HC20PW	OBSOLETE	TSSOP	PW	14		TBD	Call TI	Call TI	-40 to 85	HC20	
SN74HC20PWR	ACTIVE	TSSOP	PW	14	2000	RoHS & Green	NIPDAU SN	Level-1-260C-UNLIM	-40 to 85	HC20	Samples
SN74HC20PWT	OBSOLETE	TSSOP	PW	14		TBD	Call TI	Call TI	-40 to 85	HC20	
SNJ54HC20FK	ACTIVE	LCCC	FK	20	55	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	84039012A SNJ54HC 20FK	Samples
SNJ54HC20J	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	8403901CA SNJ54HC20J	Samples
SNJ54HC20W	ACTIVE	CFP	W	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	8403901DA SNJ54HC20W	Samples

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

⁽²⁾ **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of ≤ 1000 ppm threshold. Antimony trioxide based flame retardants must also meet the ≤ 1000 ppm threshold requirement.

⁽³⁾ MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

⁽⁴⁾ There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

⁽⁵⁾ Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

⁽⁶⁾ Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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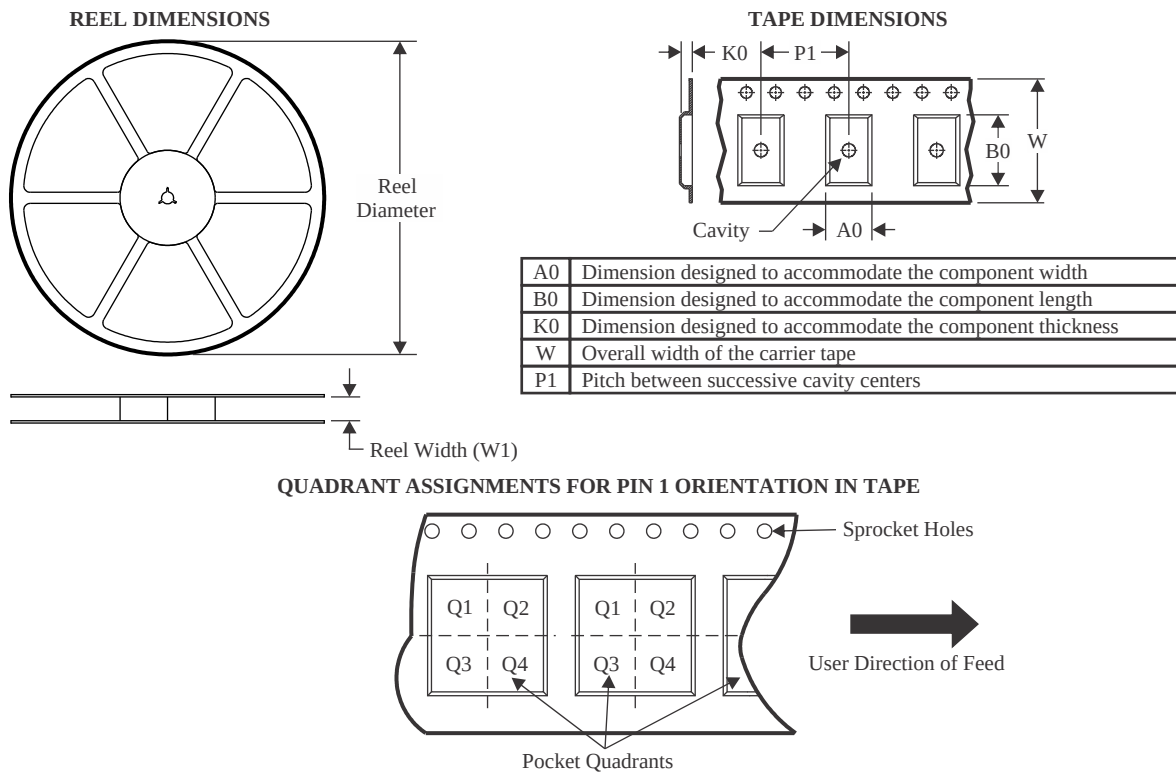
OTHER QUALIFIED VERSIONS OF SN54HC20, SN54HC20-SP, SN74HC20 :

- Catalog : [SN74HC20](#), [SN54HC20](#)
- Military : [SN54HC20](#)
- Space : [SN54HC20-SP](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications
- Space - Radiation tolerant, ceramic packaging and qualified for use in Space-based application

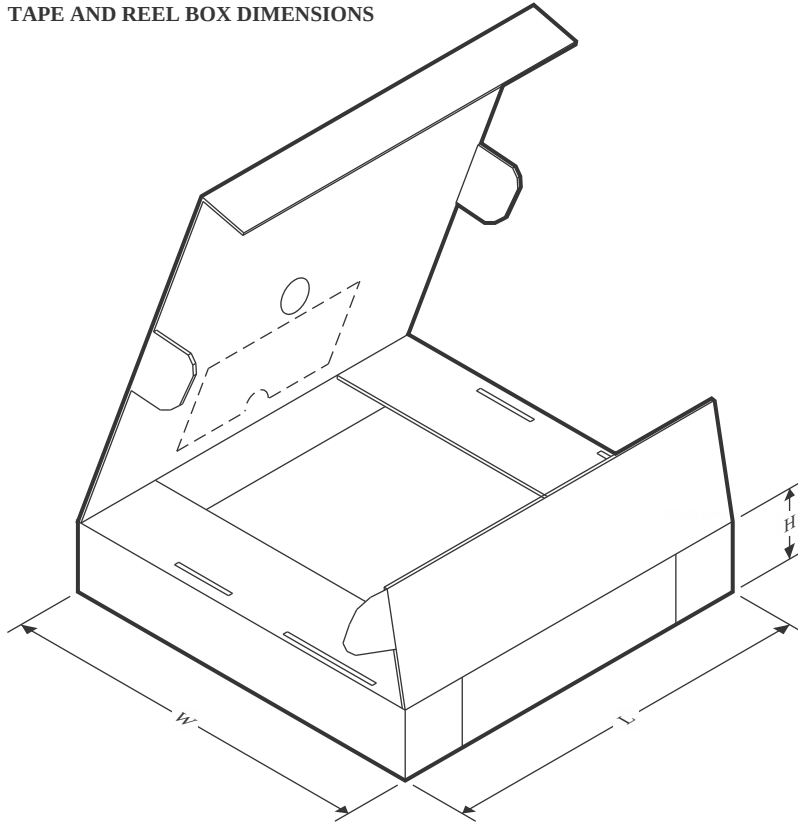
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
SN74HC20DR	SOIC	D	14	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
SN74HC20NSR	SO	NS	14	2000	330.0	16.4	8.45	10.55	2.5	12.0	16.2	Q1
SN74HC20PWR	TSSOP	PW	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1

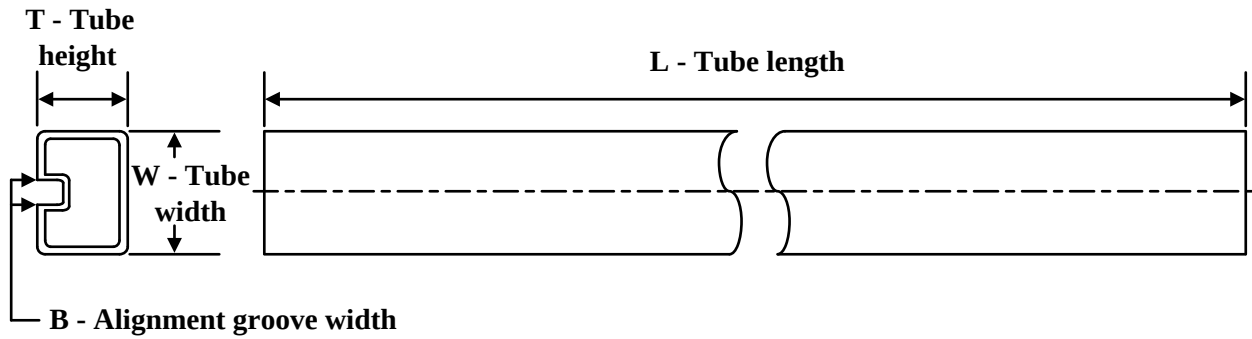
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
SN74HC20DR	SOIC	D	14	2500	367.0	367.0	38.0
SN74HC20NSR	SO	NS	14	2000	356.0	356.0	35.0
SN74HC20PWR	TSSOP	PW	14	2000	356.0	356.0	35.0

TUBE



*All dimensions are nominal

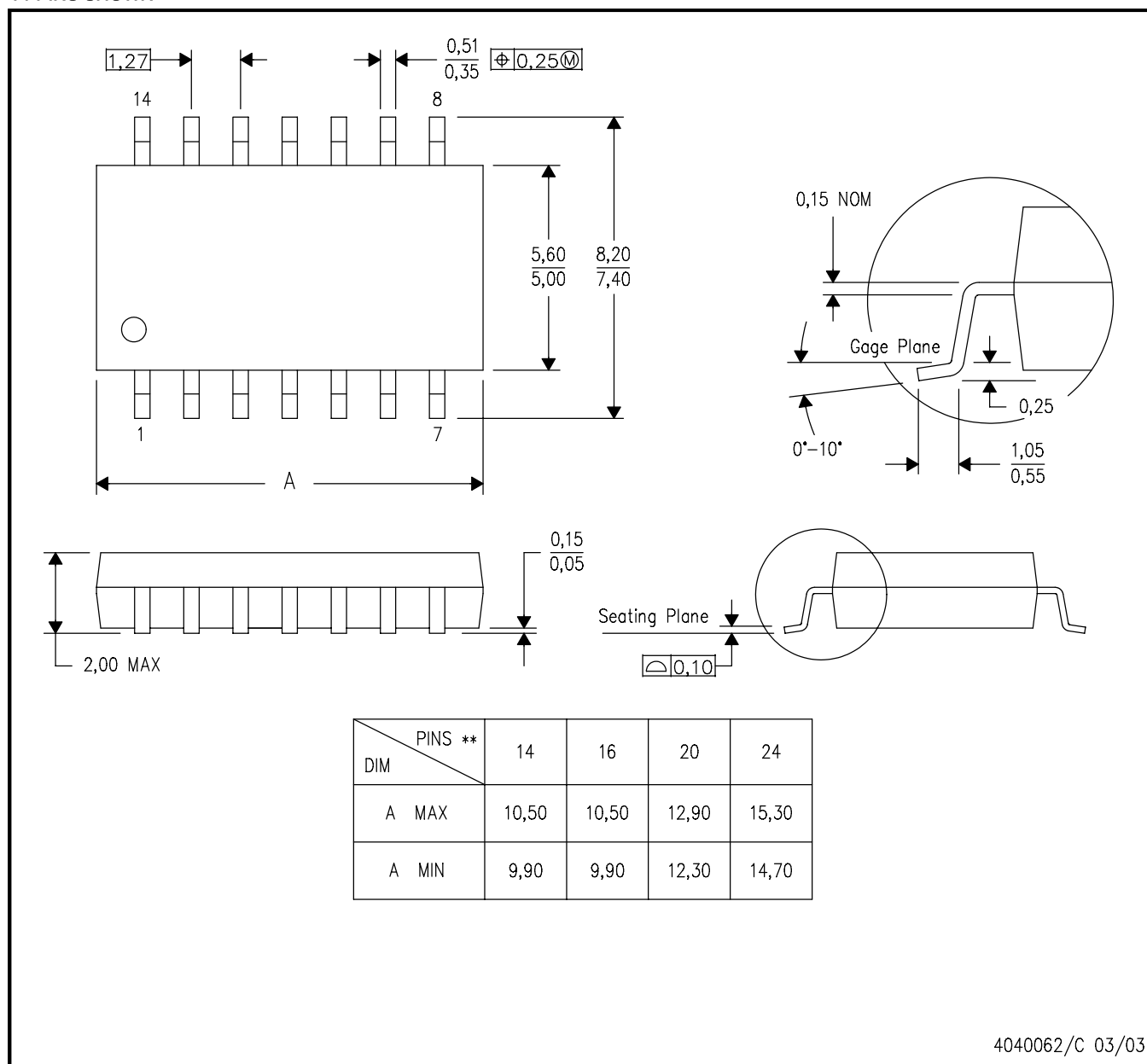
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
84039012A	FK	LCCC	20	55	506.98	12.06	2030	NA
8403901DA	W	CFP	14	25	506.98	26.16	6220	NA
SN74HC20N	N	PDIP	14	25	506	13.97	11230	4.32
SN74HC20N	N	PDIP	14	25	506	13.97	11230	4.32
SNJ54HC20FK	FK	LCCC	20	55	506.98	12.06	2030	NA
SNJ54HC20W	W	CFP	14	25	506.98	26.16	6220	NA

MECHANICAL DATA

NS (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

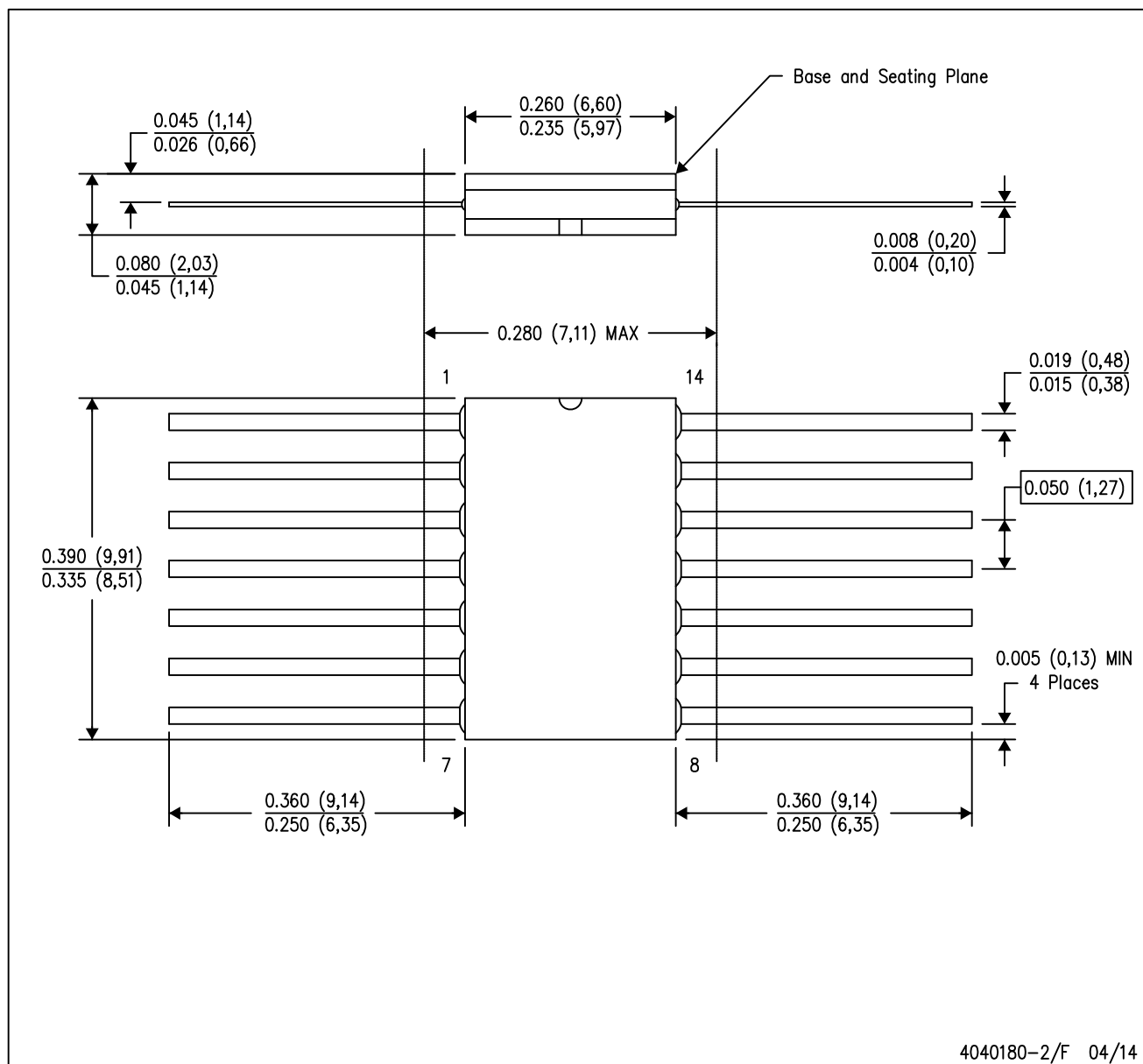
14-PINS SHOWN



- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

W (R-GDFP-F14)

CERAMIC DUAL FLATPACK



NOTES:

- All linear dimensions are in inches (millimeters).
- This drawing is subject to change without notice.
- This package can be hermetically sealed with a ceramic lid using glass frit.
- Index point is provided on cap for terminal identification only.
- Falls within MIL STD 1835 GDFP1-F14

GENERIC PACKAGE VIEW

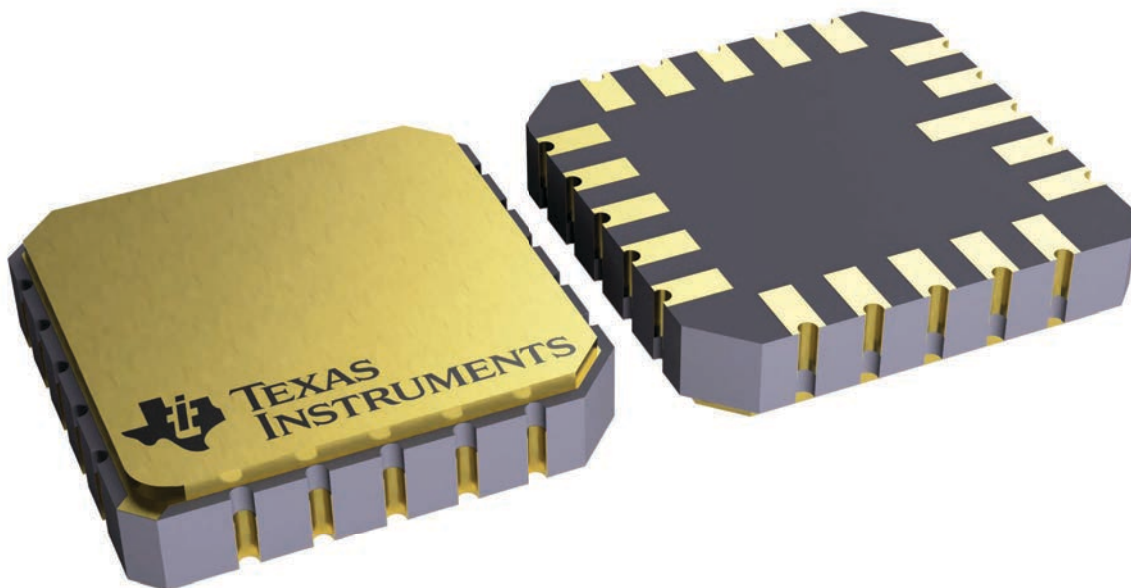
FK 20

LCCC - 2.03 mm max height

8.89 x 8.89, 1.27 mm pitch

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



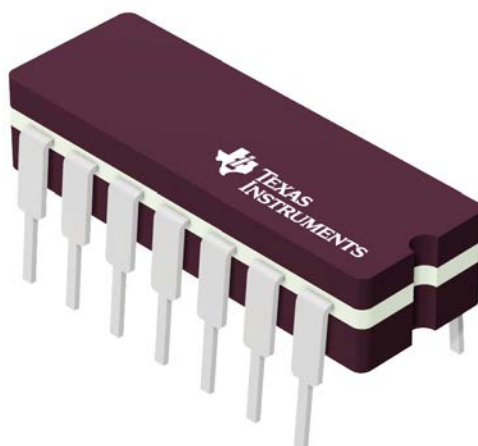
4229370VA\

J 14

GENERIC PACKAGE VIEW

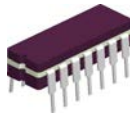
CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE

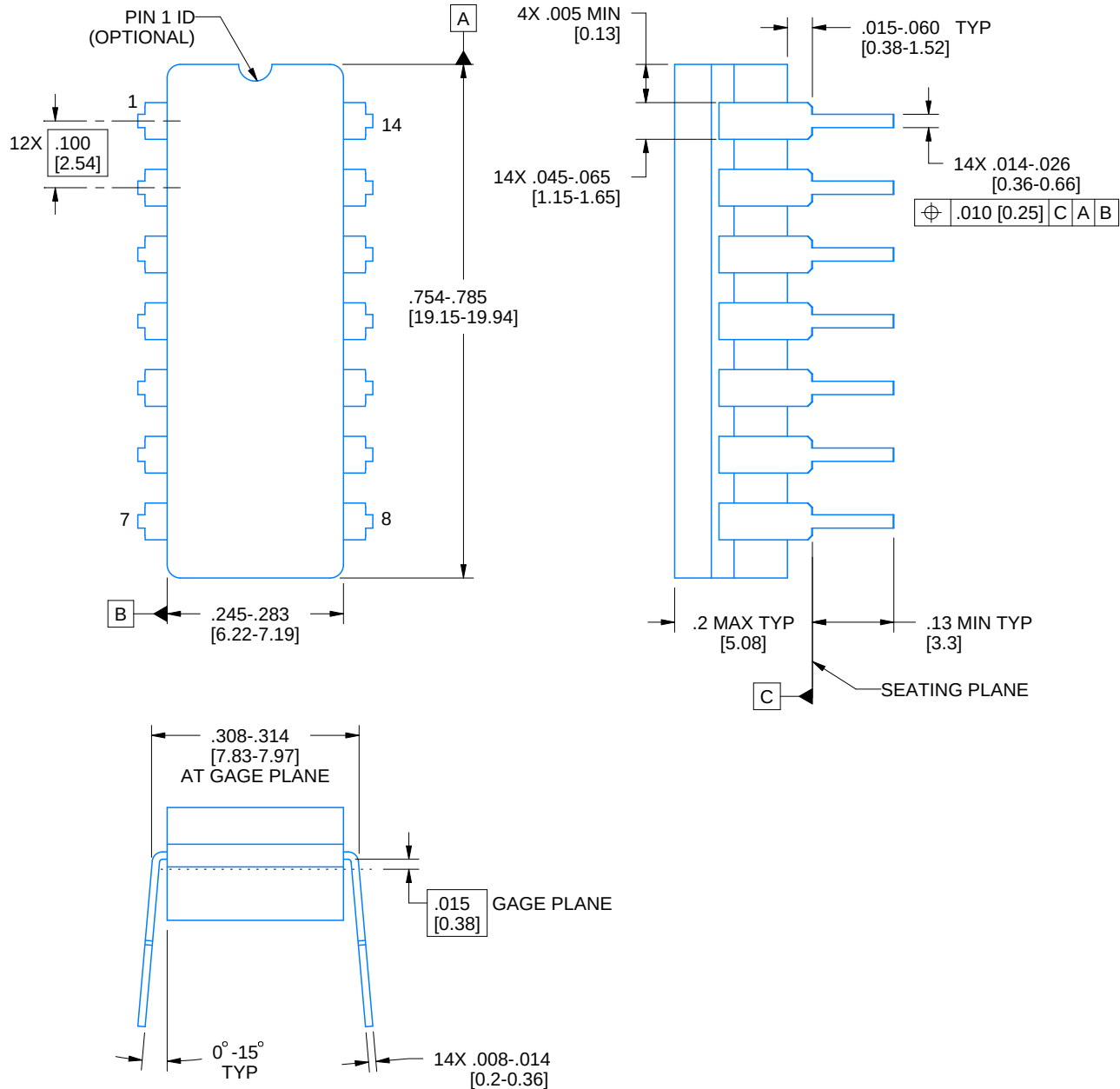


Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

4040083-5/G

J0014A**PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

NOTES:

1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.



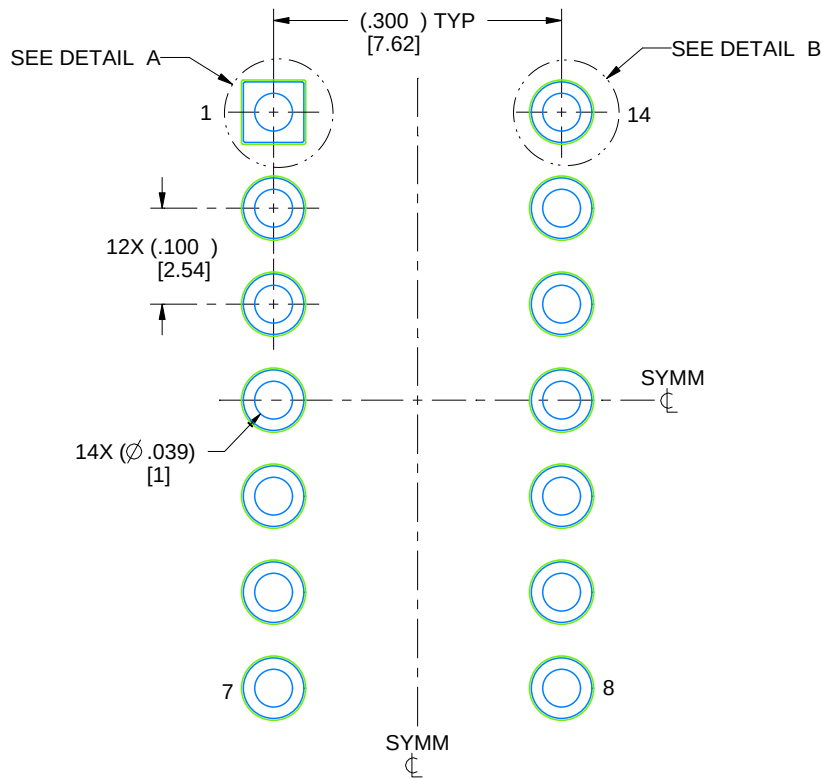
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EXAMPLE BOARD LAYOUT

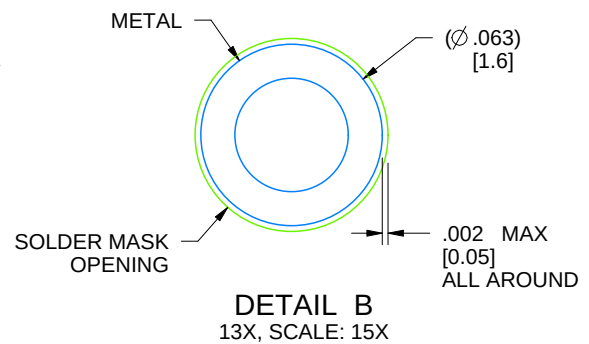
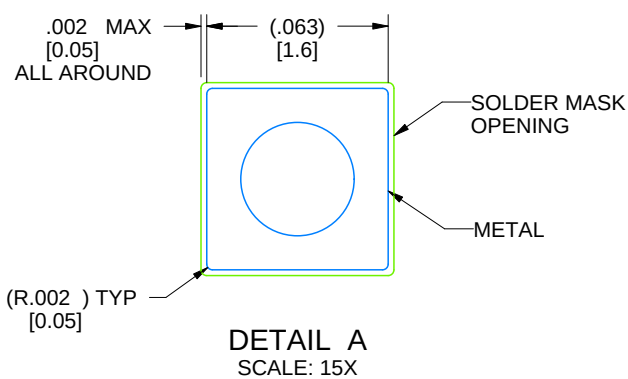
J0014A

CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE



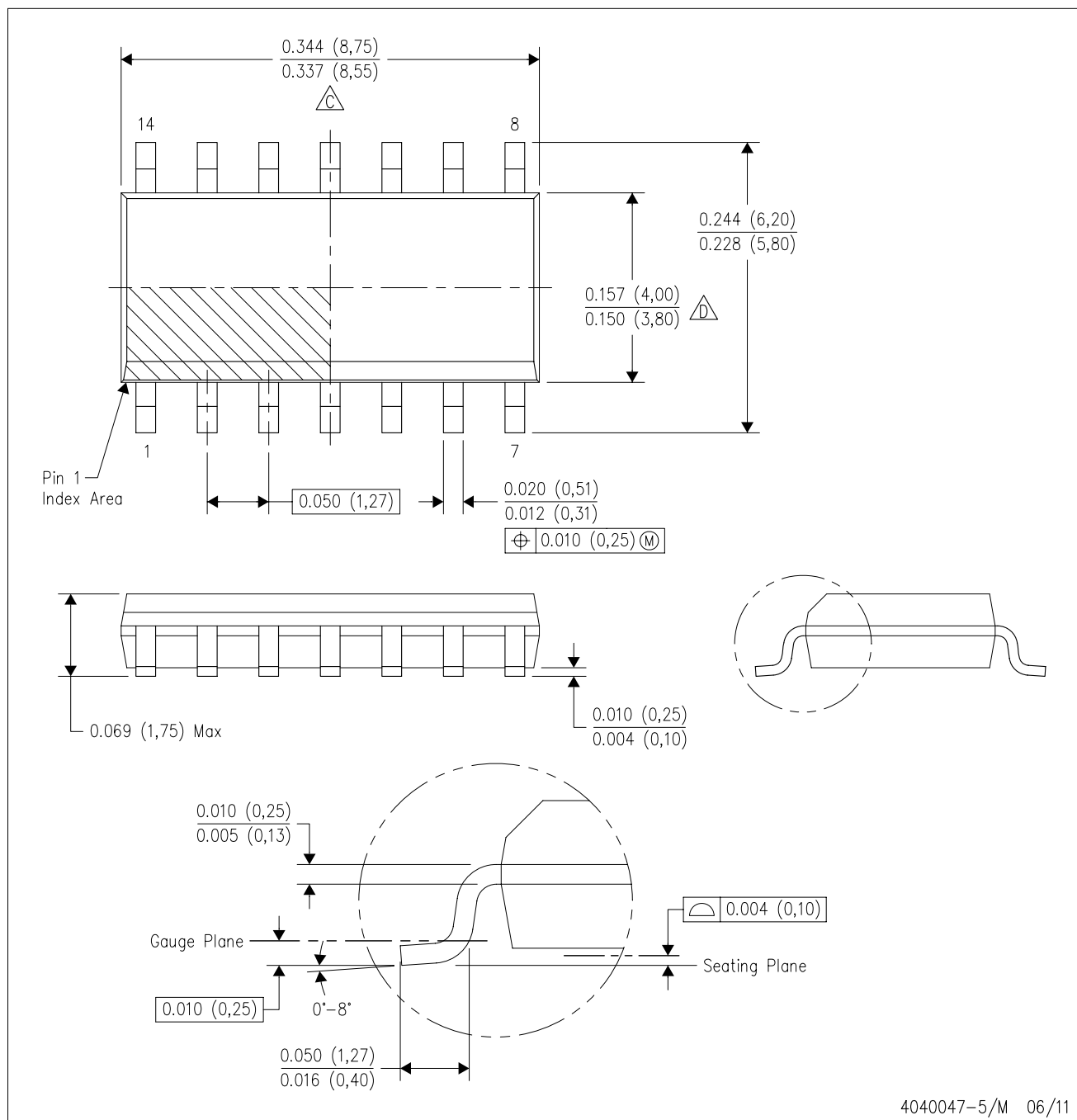
LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE: 5X



4214771/A 05/2017

D (R-PDSO-G14)

PLASTIC SMALL OUTLINE



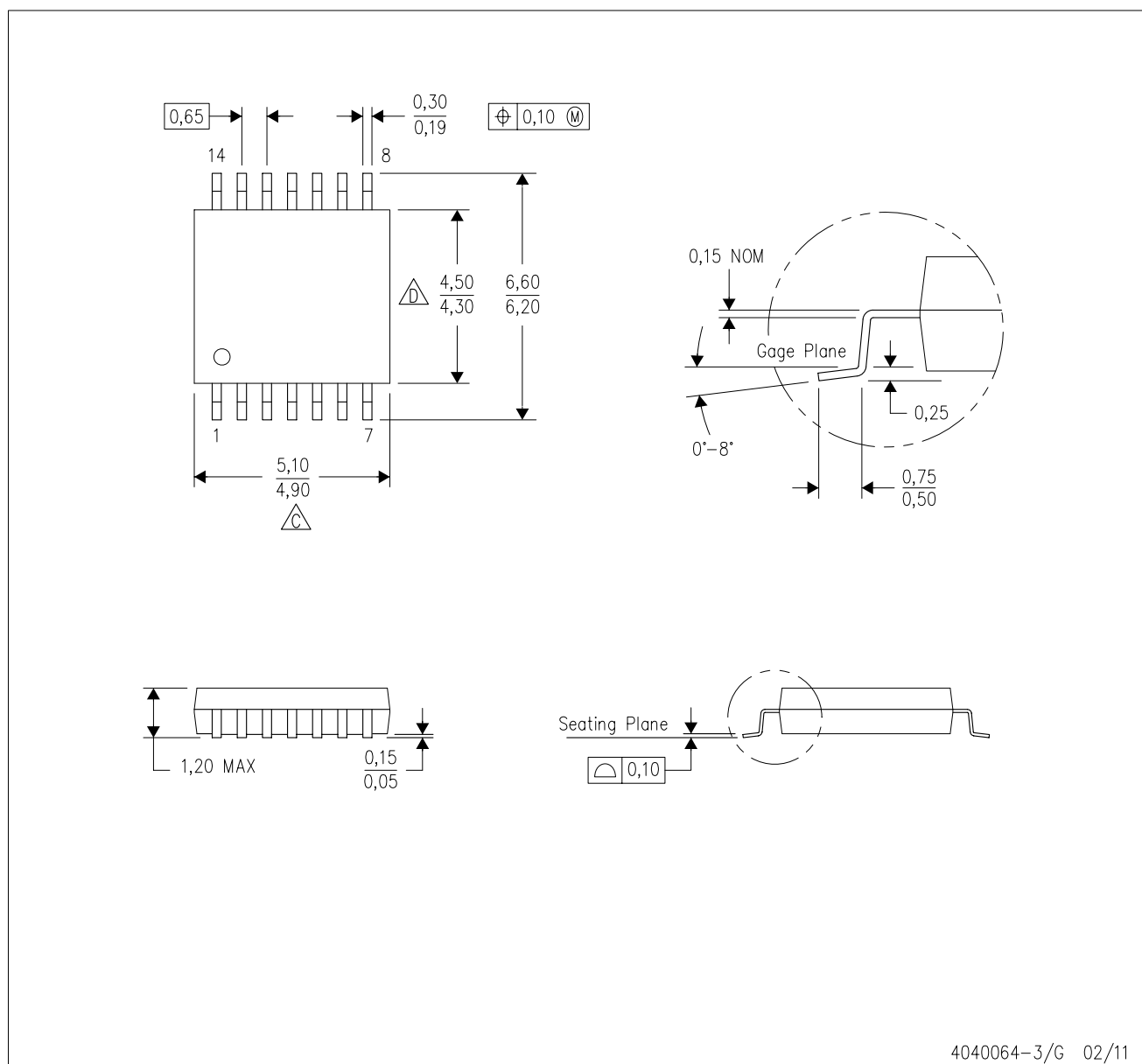
4040047-5/M 06/11

NOTES:

- A. All linear dimensions are in inches (millimeters).
- B. This drawing is subject to change without notice.
- C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
- D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
- E. Reference JEDEC MS-012 variation AB.

PW (R-PDSO-G14)

PLASTIC SMALL OUTLINE

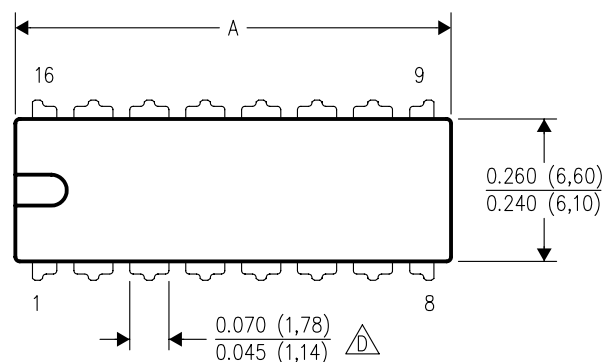


- NOTES:
- A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.
 - B. This drawing is subject to change without notice.
 - C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
 - D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
 - E. Falls within JEDEC MO-153

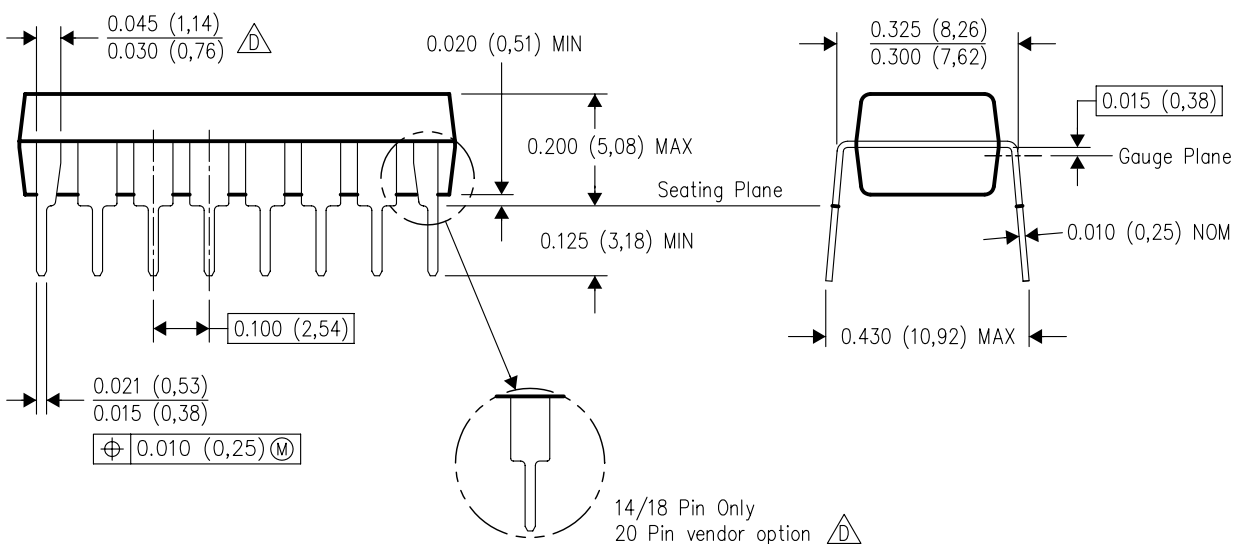
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



4040049/E 12/2002

NOTES:

- A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
-  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 The 20 pin end lead shoulder width is a vendor option, either half or full width.

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